PROJECT DOCUMENTATION

Job Market Intelligence & Skills Analytics

Project focus: Egypt

Item	Description
Team size	5 members (names to be added by the team)
Project duration	To be confirmed; timeline uses relative weeks
Main deliverable	Data collection and analysis workflow + Power BI dashboard
Optional feature	Machine-learning job-category classification

This document covers the first three required sections only. No project data has been collected yet; findings will be added after analysis.

1. Project Planning & Management

Project Idea

The project will analyze publicly available job postings relevant to Egypt to identify frequently requested skills, compare requirements across job titles and locations, examine technology combinations, and summarize salary patterns when comparable salary data are available.

Objectives

- Collect job postings from permitted public sources or APIs.
- Clean and standardize job titles, locations, and skill names.
- Analyze skill demand by role, location, and experience where data allow.
- Build an interactive Power BI dashboard and document the workflow.
- Optionally classify postings into job categories using machine learning.

Scope

Included: data collection/import, cleaning, EDA, skill analytics, dashboard, and source/limitation documentation. Excluded from the core version: a custom front-end website, a complete census of all Egyptian vacancies, and guaranteed predictions of future demand.

Timeline (proposed five-week baseline)

Phase	Schedule	Output
Scope and source review	Week 1	Approved sources and data fields
Collection and cleaning	Weeks 1–2	Raw and cleaned dataset
EDA and skills analysis	Weeks 3–4	Analysis notebook and charts
Power BI dashboard	Weeks 4–5	Interactive dashboard
Validation and final report	Week 5	Reviewed project package

Team Responsibilities

Member	Responsibility
Member 1	Project coordination, timeline, documentation compilation
Member 2	Source research and data collection pipeline
Member 3	Data cleaning, duplicate handling, skill normalization
Member 4	EDA, skill-demand analysis, salary analysis if data permit
Member 5	Power BI dashboard, testing, and visual quality

Risks and Mitigation

Risk	Mitigation
Limited access or changed source pages	Check source terms; prefer APIs/public datasets; keep a backup source.
Too few Egypt-specific postings	Pilot multiple sources; narrow roles or period and report coverage.
Duplicate or inconsistent skills	Apply documented deduplication and maintain a skill-alias dictionary.
Missing/incomparable salary data	Treat salary analysis as optional; preserve currency and pay period.
Time constraints	Prioritize the core pipeline, analysis, and dashboard; keep ML optional.

Success Metrics

- Every retained record has traceable source metadata where permitted.
- Duplicate and missing-value checks are documented.
- Skill counts can be reproduced from the cleaned dataset.
- Dashboard figures match analysis outputs and filters work correctly.
- Final report states collection period, source coverage, and limitations.

2. Literature Review

Existing Research and Solutions

ILO & CAPMAS (2024) — Labour demand and skills needs in Egypt. An Egypt-focused report examining labour demand and skills needs across occupations and sectors. It provides local context; this project differs by building a reproducible job-posting dataset and interactive dashboard. Source.

ILO (2024) — Skills mapping in Egypt. Examines skills gaps, employability, and sector/geographic issues. It informs stakeholder needs and the relevance of location-based analysis. Our project focuses more narrowly on skills mentioned in collected postings. Source.

ILO (2025) — Skills dynamics in the Arab region using online vacancy data. Uses online vacancy data to investigate skill dynamics in selected Arab countries. It supports vacancy analysis as a useful method, but its coverage should not be assumed to represent all Egyptian occupations. Our project will report its own source and sample limits. Source.

Lightcast Skills Taxonomy. Provides a structured way to standardize skills and aliases and connect skills with occupations. This motivates our normalization dictionary. Lightcast uses commercial-scale data and methods; the student project will use a smaller transparent mapping. Methodology.

O*NET OnLine. Provides structured occupational information and job categories that can inform consistent role grouping or an optional classifier. It is primarily US-oriented and is not direct evidence of Egyptian job demand. O*NET.

Comparison and Lessons

Source	Strength	Limitation	Lesson
ILO/CAPMAS	Egypt-specific context	Different research scope	Use for context, not as our results.
ILO vacancy research	Demonstrates vacancy analysis	Coverage depends on sources	Document sample and dates.
Lightcast	Consistent skills taxonomy	Commercial methods/data	Use a transparent alias dictionary.
O*NET	Structured occupation concepts	US-oriented taxonomy	Validate role labels locally.

Project Contribution

The project combines Egypt-focused context with a transparent workflow for collecting, cleaning, and analyzing a defined sample of job postings. Its practical contribution is an accessible Power BI dashboard showing skill frequency and differences by role/location where coverage permits. It will not claim to represent the entire labour market.

References

[1] ILO & CAPMAS. (2024). *Analytical report: labour demand and labour market skills needs*. Link.

[2] ILO. (2024). *Skills mapping in Egypt: Opportunity scouting and skills mapping analysis*. Link.

[3] El Hage Sleiman, S., Liepmann, H., & Mokdad, T. (2025). *Skills dynamics in the Arab region: new evidence from online vacancy data*. ILO. DOI: 10.54394/VSLG4564. Link.

[4] Lightcast. (n.d.). *Lightcast Skills Taxonomy: Methodology and Skills v10*. Link.

[5] National Center for O*NET Development. (n.d.). *O*NET OnLine*. Link.

3. Requirements Gathering

Stakeholders and Needs

Stakeholder	Need
Job seekers / students	Find commonly requested skills for a target role.
Career centres / educators	Understand recurring skill themes for career guidance.
HR / recruiters	Compare requirements across roles and locations.
Researchers / workforce organizations	Review transparent, documented skill-demand evidence.
Project team / supervisor	Verify data quality, calculations, and deliverables.

Functional Requirements

• Collect/import public job postings from documented sources relevant to Egypt.
• Extract available fields: title, location, skills, experience, industry, salary, posting date, and source.
• Identify duplicates, profile missing values, and standardize common skill-name variants.
• Rank skills by the number of distinct postings that mention them.
• Compare skills by job title and location; explore technology combinations.
• Analyze salaries only when currency and pay period are comparable.
• Build a Power BI dashboard with filters for available role, location, and experience fields.
• Optional: classify postings into predefined job categories using an evaluated ML model.

Non-Functional Requirements

Quality	Requirement
Performance	Dashboard should remain responsive for the final dataset; set a numeric target after testing.
Accuracy / data quality	Document cleaning rules and reconcile dashboard figures with the dataset.
Usability	Readable labels, clear filters, consistent units, and simple navigation.
Security / privacy	Do not collect applicant personal data; protect API keys and credentials.
Reliability	Log collection errors and handle missing or malformed records safely.
Maintainability	Separate collection, cleaning, analysis, and dashboard steps; document code.
Compliance	Respect source terms, access rules, rate limits, and applicable licenses.

Core Data Fields

Job title; location; skills (raw and normalized); experience; industry; salary range/currency/pay period where available; posting date; source name/URL or identifier; and collection date. Missing values must remain unknown rather than being treated as zero.

Dashboard Requirements

The dashboard should include an overview of the dataset, top skills, skills by role, location analysis, experience distribution, salary analysis where reliable, and common skill combinations. Filters should work across visuals, and the dashboard should state the collection period and data limitations.

Acceptance Criteria

• Source and collection method are documented and permitted.
• Cleaning, duplicate rules, and skill normalization are reproducible.
• Skill rankings can be recalculated from the cleaned dataset.
• Dashboard filters work and key figures match the analysis.
• The report explains sample coverage and avoids generalizing beyond the collected postings.

Assumptions to Confirm

The actual deadline, data sources, selected job categories, and target dataset size remain to be confirmed with the supervisor. The five-week timeline is only a proposed baseline. Machine learning and salary analysis depend on data quality and available time.